

LAB SESSION: INTRODUCTION TO GIT, GITHUB, AND TESTING

Lab objective

In this practical session, you'll get hands-on experience using Git and GitHub. You'll learn how to create branches, commit changes, push code, perform unit tests with pytest, and verify your code with linting. You will also practice the pull-request (PR) workflow commonly used in collaborative software development.

Preparation and Setup

- **Install GitHub Desktop** (GitHub client):
 - <https://github.com/apps/desktop>
- **Create a GitHub account** and configure authentication in your GitHub Desktop client.

Forking and Cloning the Repository

- **Fork** the following repository:
 - <https://github.com/Hermann-web/learn-git-pytest>
- **Clone your forked repository** to your local machine:
 - Open GitHub Desktop.
 - Select your forked repository, then click on "**Clone**".

Practical Exercises Workflow

To familiarize yourself with GitHub collaboration, follow the steps below to solve each exercise:

Step-by-step Procedure

1. **Create a new branch** for each exercise with the naming convention:

exo I/firstname-lastname

Replace firstname and lastname with your actual first and last names.

2. **Solve the exercise by writing the code.**
3. **Run unit tests locally** to ensure your solution works as expected:

pytest tests/

4. **Commit your changes** regularly with clear commit messages.
5. **Push your branch** to your forked repository using GitHub Desktop.

6. **Create a Pull Request** from your branch (exo1/firstname-lastname) towards the main repository.
7. **Ensure GitHub Actions passes:**
 - Check if your unit tests run successfully on GitHub Actions.
 - Ensure your code complies with linting rules.
8. **Repeat this procedure** for exercises exo2 and exo3.

Important Notes:

- Always verify that your unit tests (pytest) pass locally before pushing.
- Ensure that your code follows linting guidelines to maintain good coding standards.
- Make sure GitHub Actions confirms successful testing and linting for your PR.

Summary of Workflow (for quick reference)

- Fork → Clone → Create branch → Code → Test (pytest) → Commit → Push → PR → Verify GitHub Actions

Good luck and happy coding!

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